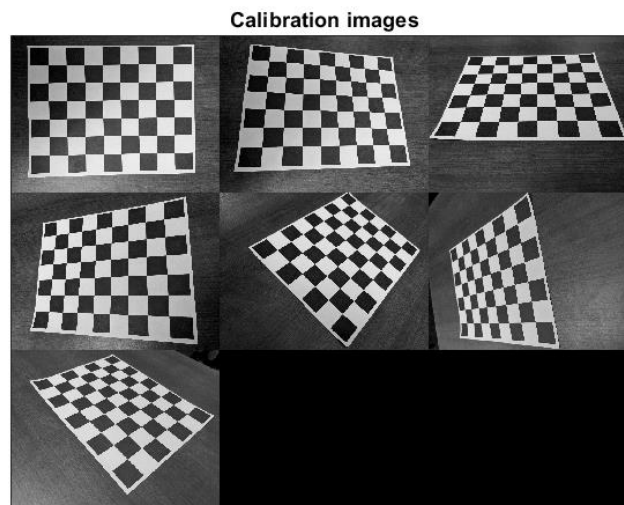


Camera Mosaic

Introduction

The report explores the creation of camera mosaics using Harris Corner Detection, a feature detection algorithm critical for image stitching. The process begins with precise camera calibration to establish accurate intrinsic parameters, ensuring reliable image processing. We examine the effectiveness of Harris Corner Detection across different scenarios, including the stitching of images from an LSC building mural, a cinder brick wall, and a Whole Foods mural, using varying overlap percentages. This approach provides insights into the algorithm's strengths and limitations in diverse imaging contexts. Camera calibration plays a pivotal role in enhancing the precision of feature detection and alignment, while the choice of images and overlap percentages offers a comprehensive understanding of the algorithm's performance in the real-world.

Camera Calibration



The camera calibration process was conducted on a Google Pixel 5a using the Caltech Camera Calibration Toolbox. The calibration was achieved by analyzing seven images of a checkerboard taken from different angles. The purpose of this exercise was to precisely determine the camera's intrinsic parameters, which are crucial for accurate image processing and computer vision applications.

Calibration parameters after initialization:

```
Focal Length:      fc = [ 3284.44524  3284.44524 ]
Principal point:    cc = [ 2015.50000  1511.50000 ]
Skew:              alpha_c = [ 0.00000 ] => angle of pixel = 90.00000 degrees
Distortion:        kc = [ 0.00000  0.00000  0.00000  0.00000  0.00000 ]
```

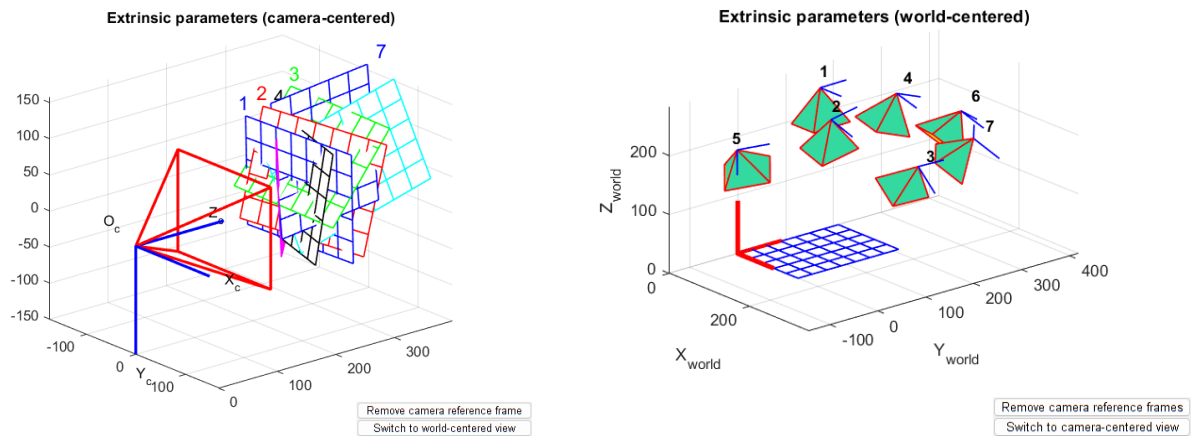
Main calibration optimization procedure - Number of images: 7

```
Gradient descent iterations: 1...2...3...4...5...6...7...8...9...10...11...12...13...14...15...16...17...18...19...done
Estimation of uncertainties...done
```

Calibration results after optimization (with uncertainties):

```
Focal Length:      fc = [ 3207.30462  3209.95559 ] +/- [ 24.04616  22.78221 ]
Principal point:    cc = [ 1992.72914  1557.30514 ] +/- [ 17.06089  19.39834 ]
Skew:              alpha_c = [ 0.00000 ] +/- [ 0.00000 ] => angle of pixel axes = 90.00000 +/- 0.00000 degrees
Distortion:        kc = [ 0.10949  -0.35021  0.00450  -0.00033  0.00000 ] +/- [ 0.02295  0.08895  0.00184  0.00209  0.00000 ]
Pixel error:       err = [ 1.68709  1.41166 ]
```

Note: The numerical errors are approximately three times the standard deviations (for reference).



Calibration Parameters

a. Focal Length

Focal length(f_c) is one of the most crucial parameters in camera calibration. It is represented in pixels rather than physical units. Initially, the calibration showed a focal length of 3284.44524 in both the x and y dimensions, indicating a very high zoom level. After optimization, it slightly reduced to 3207.30462 (x-dimension) and 3209.95559 (y-dimension). This change suggests that the camera's lens has a high magnification capability, and the optimization helped in refining its accuracy. The focal length directly affects the field of view of the camera; a higher focal length results in a more zoomed-in view.

b. Principal Point

The principal point(c_c) is the point on the image sensor where the optical axis intersects it. Initially, it was [2015.50000, 1511.50000] and changed to [1992.72914, 1557.30514] after optimization. This shift indicates a slight adjustment in the alignment of the sensor and the lens. The principal point is crucial for understanding how the camera's lens and sensor are aligned, affecting image centering and perspectives. A correctly aligned principal point ensures that images are centered, and perspectives are accurately rendered.

c. Skew

The skew coefficient(α_c) represents the angle between x and y pixel axes. Here it has remained consistent at 0.00000 before and after calibration, indicating that the pixel axes are perfectly orthogonal at 90 degrees. This is an ideal scenario as it confirms the absence of skew distortion, ensuring that the geometrical integrity of the images is maintained, and the pixel grid is well aligned.

d. Distortion

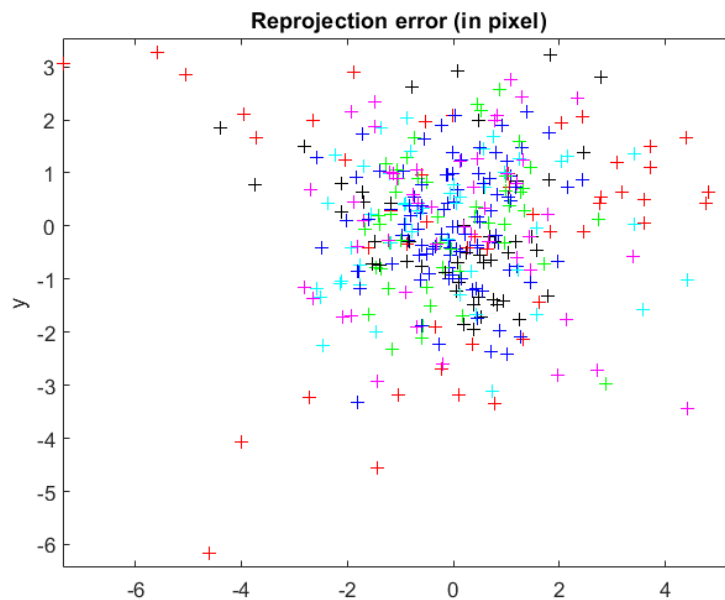
The distortion(k_c) parameters are critical in understanding lens imperfections. They include radial and tangential distortion coefficients. Initially, all coefficients were zero, meaning no distortion was assumed. However, after optimization, the values changed significantly, especially the first two radial distortion coefficients [0.10949 and -0.35021]. These changes indicate that the lens does exhibit some distortion, which the calibration process has identified and quantified. Lens distortions can warp the geometry of captured images, and these adjustments help in accurately rectifying such distortions.

e. Pixel Error

The pixel error(err) provides an estimate of the accuracy of the calibration. The values [1.68709, 1.41166] represent the average deviation in pixels for the x and y coordinates, respectively.

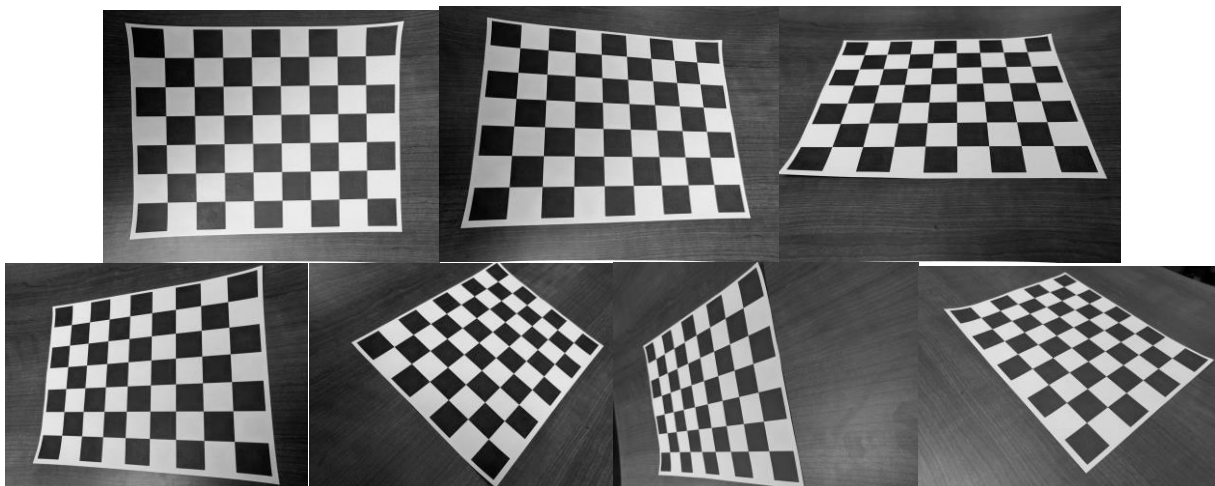
Considering the original image resolution of 4032x3024 pixels, these low error values demonstrate a high accuracy level in the calibration process.

The reprojection error graph, which shows points clustered within a ± 3 pixel range, is a testament to the consistent accuracy of the calibration. This graph visualizes the difference between the observed image points and those predicted by the calibrated camera model. The tight clustering within this range, especially with the high-resolution images used, underscores the reliability and robustness of the calibration. It indicates that the camera, post-calibration, will perform with high consistency and accuracy.



The calibration results are instrumental for applications requiring high precision in image capture and analysis. The refined focal length and principal point ensure that images are captured with precise zoom and alignment, essential for tasks like augmented reality, where digital and real-world images are merged. The absence of skew and the correction of lens distortion enhance the overall image quality, reducing artifacts and improving the accuracy of feature detection and mapping. Moreover, the low pixel error and tight reprojection error range indicate the reliability and consistency of the camera's performance post-calibration.

Undistorted Images

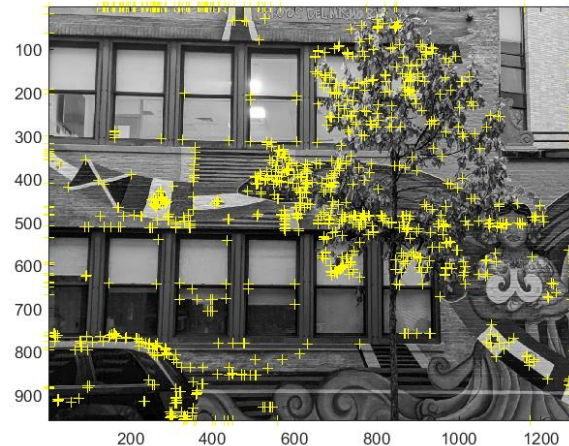
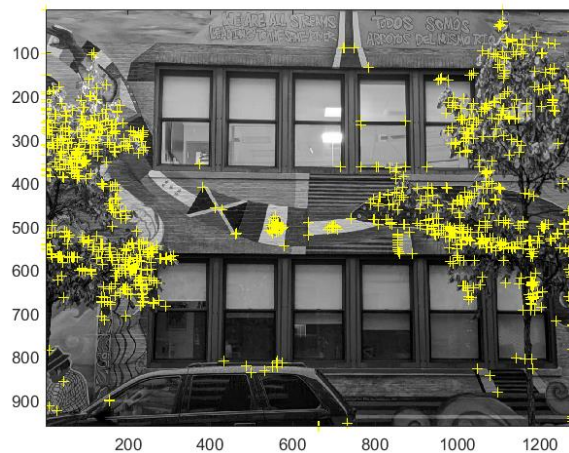


LSC Mosaic

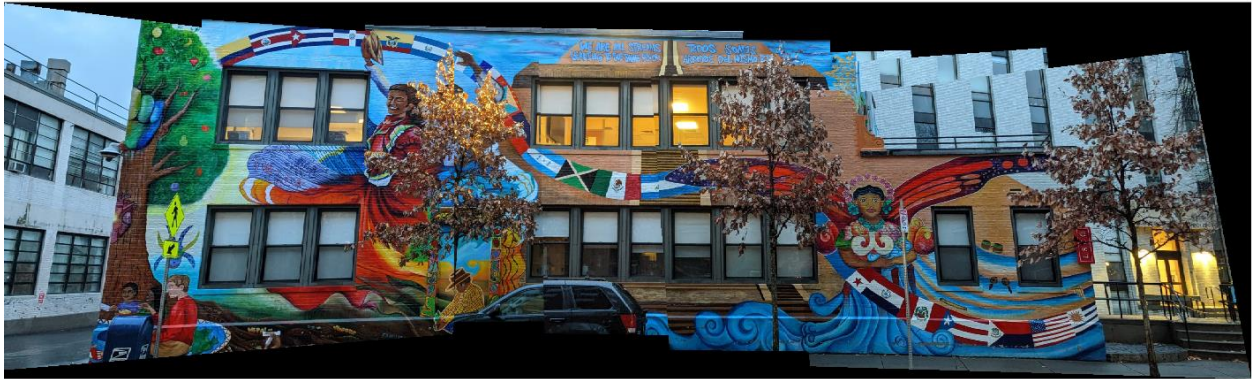
I took seven images of the LSC building mural to create a panoramic view. Nowadays, phone cameras have evolved, due to which they already provide us the undistorted images, and if we try to undistort image using the toolbox it will make the images even more distorted and bending towards the corner. For this process, I employed the Harris Corner Detection technique, specifying 1000 features with a [2x2] tile distribution. This approach was aimed at ensuring a thorough feature distribution across each image, enhancing the alignment accuracy in the stitched panorama.



The resulting panorama successfully captures the mural with precise alignment, demonstrating the effectiveness of the Harris Corner Detection in identifying and utilizing key features of the building. The mural, characterized by distinct edges and corners, was perfectly joined, showcasing the strength of the corner detection method in handling well-defined structures.

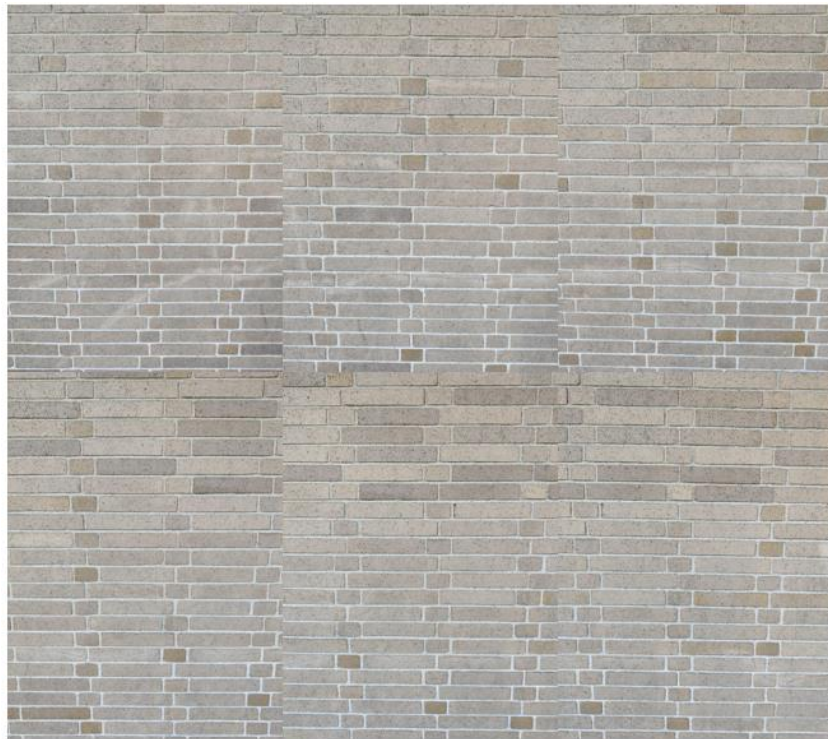


However, the panorama exhibited slight misalignments in areas with less defined features. Notably, a small pathway painted between the windows and a car on the road were not perfectly aligned. This misalignment can be attributed to the inherent limitations of the Harris Corner Detection technique, which excels in identifying distinct corners but may falter with less prominent features or in areas where the 3D structure introduces complexity. In the case of the pathway and the car, these elements lack the pronounced corners and edges that the algorithm primarily targets, leading to the minor alignment issues observed.

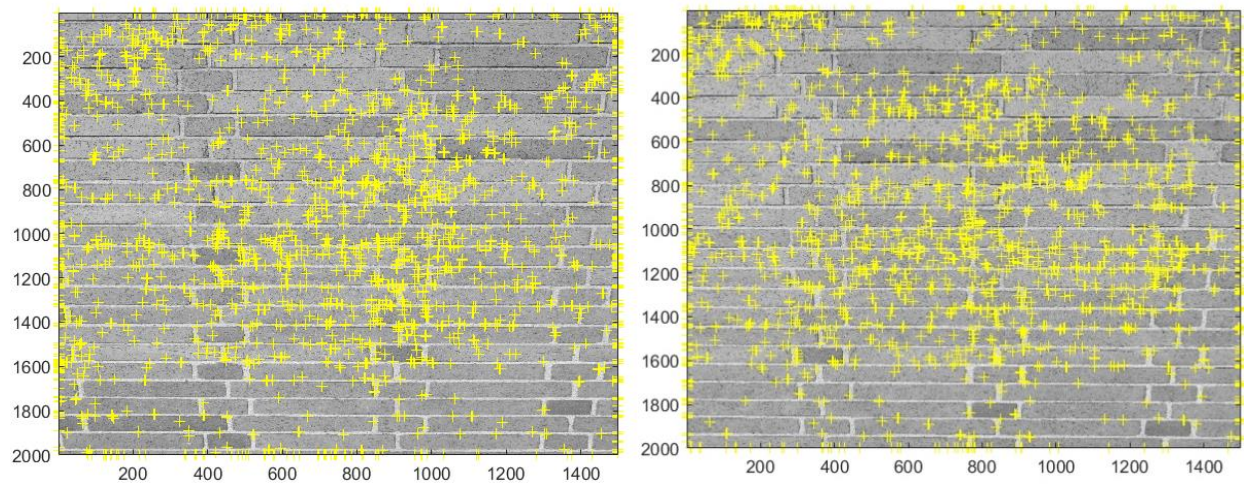


Cinder brick Wall

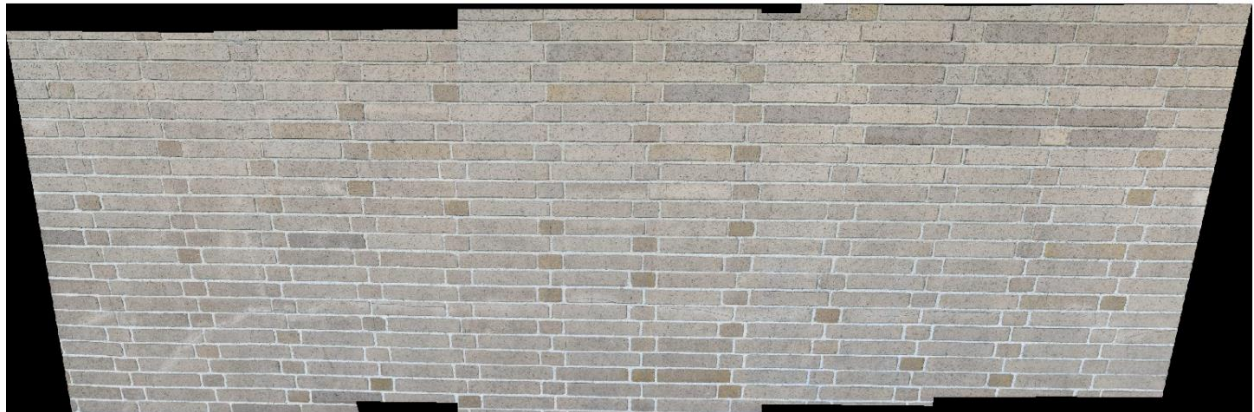
I captured a series of six images, each overlapping by 50%, of a cinder brick wall near Cabot Centre. The wall displayed a regular and repetitive pattern, which presented a unique challenge for feature detection and image stitching.



To address this, I employed the Harris Corner Detector, setting it to detect 2,000 features with a [2x2] tile approach. This configuration was chosen to ensure an even distribution of features across the images, providing effective non-maximal suppression of features.



The majority of the detected features were concentrated around the seams and corners of the bricks, as shown in the figure. This distribution was expected, given the numerous distinct corners present in the cinder brick pattern. The higher number of Harris features, compared to previous experiments, was necessary due to the repetitive nature of the brick pattern, which typically poses challenges in feature-based image stitching. Despite these challenges, the results were impressive. The stitched panorama showed minimal distortion, particularly towards the borders, indicating a high degree of alignment. The increase in Harris features played a crucial role in achieving near-perfect stitching, demonstrating the adaptability of the technique to different types of structures and patterns.



LSC Mosaic vs Cinder brick Wall

Comparing the results of the LSC building mural and the cinder brick wall provides insightful observations about the Harris Corner Detection technique's performance. In the case of the LSC mural, the algorithm excelled in aligning well-defined structures like the building's corners and edges, resulting in a perfectly joined mural. However, it faced minor challenges in areas with less pronounced features, like the pathway and car.

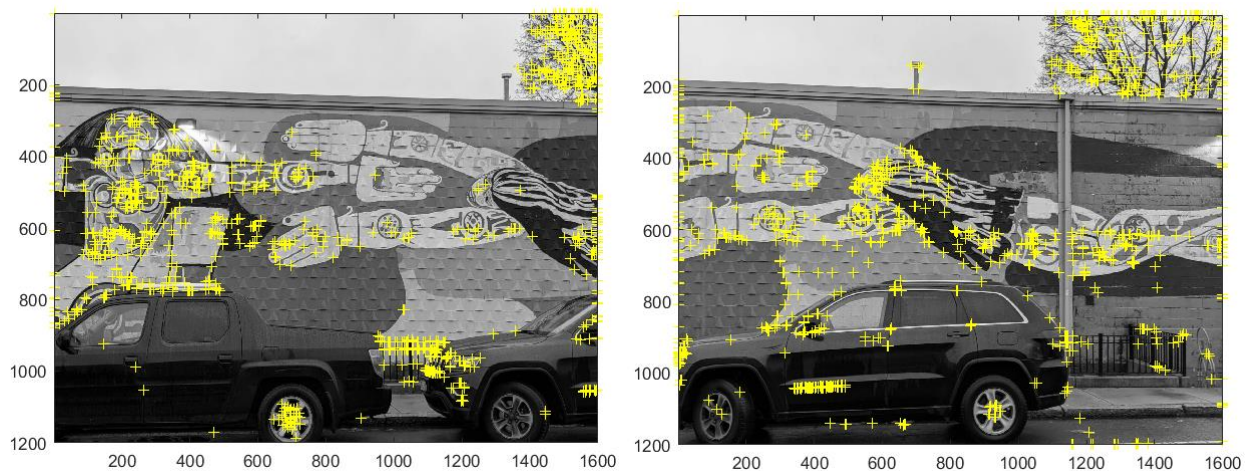
On the other hand, the cinder brick wall, characterized by its repetitive pattern, required a higher number of Harris features (2,000 compared to 1,000 in the LSC mural). This adjustment was crucial in handling the uniformity and repetitiveness of the brick pattern, which can often confuse feature detection algorithms. The successful alignment of the brick wall panorama, with minimal distortion even at the borders, highlights the Harris Corner Detection's capability to adapt to different textural complexities by adjusting the number

of features. This comparison underscores the importance of tailoring the feature detection parameters to the specific characteristics of the subject being photographed, ensuring optimal results in diverse photographic scenarios.

Mural 50% Overlap

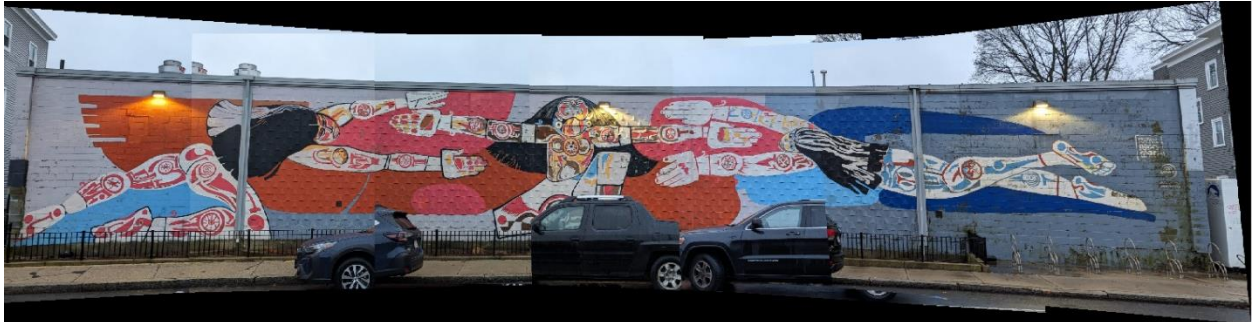


In the first scenario, capturing the mural at Whole Foods on Centre Street, I utilized a 50% overlap strategy, taking six images. This methodology, combined with 1000 Harris features and a [2x2] tile distribution, detected the edges of the mural and the building corners properly.



This was highly effective in seamlessly stitching the mural. The images aligned perfectly, showcasing the continuity of the mural with great precision. However, challenges arose in representing parked cars on the road. Despite the cars being stationary, the 50% overlap resulted in them appearing with their halves cut off

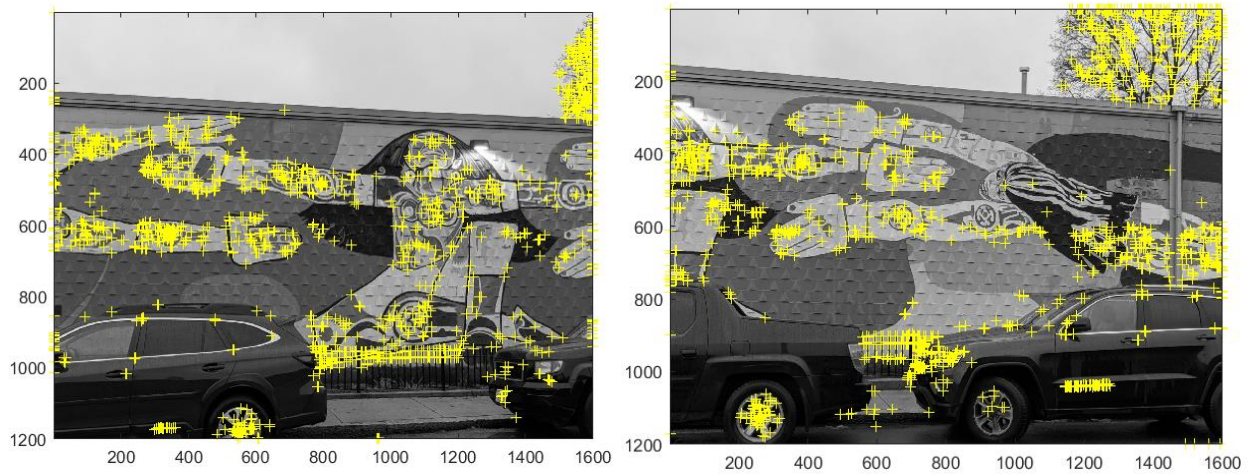
in the stitched panorama. This outcome highlights a limitation in handling objects at the periphery of overlapping images. Conversely, static elements like the tree at the corner of the building were captured almost perfectly, demonstrating the technique's effectiveness with stationary objects and consistent backgrounds.



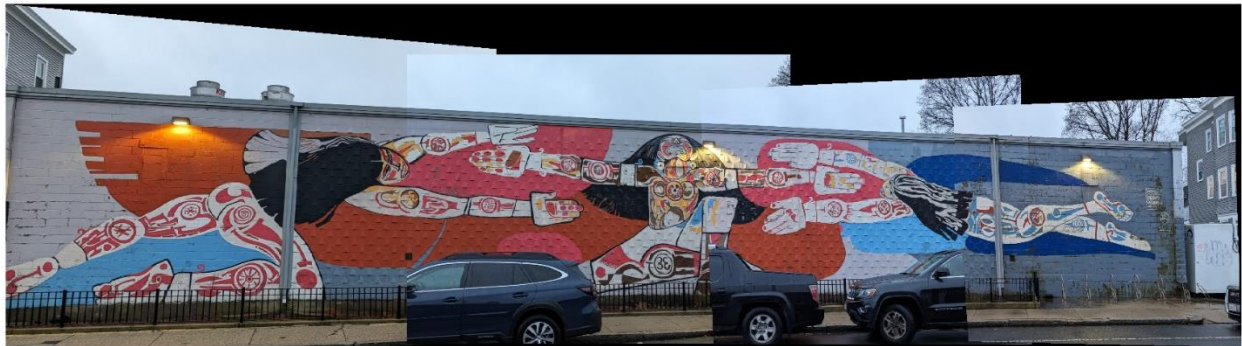
Mural 15% Overlap

In the second set, with a reduced overlap of 15% and only four images, the Harris feature count was increased to 1500, maintaining the [2x2] tile distribution. The edges were detected in the grills and the mural and majorly spread across the hands of the mural.





This setup also achieved a near-perfect alignment of the mural. However, the lower overlap significantly affected the representation of the parked cars and the 3D tree. The cars were more noticeably distorted, with their halves appearing more prominently cut off. The 3D tree, located at the edge of the mural, was fragmented and replicated across the panorama, appearing disjointed and in small chunks. This effect can be attributed to the reduced overlap not capturing enough common features between adjacent images, leading to these alignment challenges.



Overlap Comparison

Comparing the two approaches provides valuable insights into the impact of overlap percentage in panorama stitching. The 50% overlap was more effective in maintaining continuity in the mural and the static environment, but it presented limitations in accurately stitching objects at the edges of overlapping frames, like the parked cars. In contrast, the 15% overlap, despite a higher feature count, led to more pronounced distortions and misalignments, particularly noticeable in the fragmented appearance of the parked cars and the 3D tree.

This comparison underscores the importance of a balanced overlap percentage in capturing both the central subject and peripheral elements in a scene. While higher overlaps are preferable for seamless central subjects, they can introduce challenges in stitching edge elements. Conversely, lower overlaps, although reducing image redundancy, can struggle with accurately aligning features in areas with less overlap. Therefore, the choice of overlap percentage should be carefully considered, particularly in scenes with important elements at the edges of the frame. Adjusting the feature count and distribution strategy can mitigate some issues, but the inherent trade-offs between different overlap percentages remain a crucial consideration in panorama stitching.

Conclusion

The experiments conducted, ranging from camera calibration to stitching different scenes, highlight the nuanced capabilities of the Harris Corner Detection algorithm in creating image mosaics. While it excels in aligning well-defined features, challenges arise in areas with less distinct characteristics. The varying overlap percentages in the Whole Foods mural demonstrate how this choice impacts the stitching quality, especially for peripheral elements. These findings emphasize the importance of parameter selection in panorama stitching and suggest areas for further improvement in adaptive feature detection for diverse and dynamic scenes. The results also illustrate the critical balance between feature detection precision and the adaptability of stitching techniques to different environments, paving the way for enhanced methodologies in panoramic imaging.